PRISMA 2020 Eligibility Stage: Quality Criteria Scoring — Section 5.1 (Online and Incremental Learners)

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1. Scope of This Scoring Pass

This report extends the Step 4 eligibility scoring to the full candidate set retrieved for Section 5.1 (Online and Incremental Learners) of the review, comprising all 12 full-text articles retained for this subsection following Step 3 screening. Each article was scored honestly against its full text rather than against title and abstract alone.

QC1 (problem statement and scope clarity) was scored against fit to this review's Section 5.1 scope specifically, namely learner architectures for intrusion detection whose update mechanism can, or explicitly cannot, be conditioned on a typed diagnosis of drift, rather than against general online-learning-under-drift relevance alone, and rather than against the clarity of the problem statement in the article's own, different research question. An article can present a clear problem statement for a different question, such as unsupervised feature selection, class-imbalance handling, or active-learning budget allocation, and still score 0 on QC1 here if that problem statement is not the one this subsection is screening for.

QC4 (empirical extraction detail) was scored as 0 wherever an article's full text contained no drift-type, distributional-component, or adaptation-mechanism data extractable into this review's own extraction fields, regardless of how much extraction detail the article offered for its own, different research question. This reflects a consistent, review-specific application of the criterion rather than a per-article adjustment.

QC2 (methodological transparency) and QC3 (venue indexing and citability) were scored on their ordinary, article-intrinsic basis, since these criteria measure the rigor and citability of the work itself rather than its fit to this particular review.

2. Quality Assessment Criteria Applied

Table 1 reports the four quality criteria (QC1 to QC4) applied in this eligibility round, together with the scoring scheme used for each. A quality score is computed as the sum of applicable criterion scores divided by the number of applicable criteria, scaled to a 0 to 4 range.

Table 1. Quality assessment criteria applied to Section 5.1 candidate articles

Code	Criterion	Scoring
QC1	Problem statement and scope clarity	Yes (1) / Partial (0.5) / No (0)

Code	Criterion	Scoring
QC2	Methodological transparency	Yes (1) / Partial (0.5) / No (0)
QC3	Venue indexing and citability	Yes (1) / Partial (0.5) / No (0)
QC4	Empirical extraction detail (where applicable)	Yes (1) / Partial (0.5) / No (0) / N/A

Articles published in 2020 or later were retained where their total quality score met or exceeded 2.5 out of 4. Articles published before 2020 were retained where their total quality score met or exceeded 1.5 out of 4.

3. Results

Of the 12 full-text articles assessed for Section 5.1 against the quality criteria in Section 2, 8 articles met or exceeded their applicable threshold and were retained, while 4 articles fell below their applicable threshold and were excluded. Table 2 lists each assessed article together with its per-criterion and total quality scores, applicable threshold, and eligibility decision, sorted with included articles listed first, followed by excluded articles.

Table 2. Quality criteria scoring results for Section 5.1 candidate articles, sorted by decision

No.	Citation	Year	QC1	QC2	QC3	QC4	Total	Threshold	Decision
1	A Survey of Data Stream-Based Intrusion Detection Systems (2025)	2025	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
2	A comparative study on online machine learning techniques for network traffic streams analysis (2022)	2022	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
3	Evolving Fuzzy-Rule-Based Classifiers From Data Streams (2008)	2008	1.0	0.5	1.0	0.5	**3.0**	≥1.5	**Include**
4	Incremental on-line learning: A review and comparison of state of the art algorithms (2018)	2018	1.0	1.0	1.0	1.0	**4.0**	≥1.5	**Include**
5	An Incremental Learning Method Based on Dynamic Ensemble RVM for Intrusion Detection (2022)	2022	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
6	IL-IDS: An incremental learning approach with confined data streams for intrusion detection (2025)	2025	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
7	Concept Drift-Based Intrusion Detection For Evolving Data Stream Classification In IDS: Approaches And Comparative Study (2024)	2024	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**

No.	Citation	Year	QC1	QC2	QC3	QC4	Total	Threshold	Decision
8	The Impact of Diversity on Online Ensemble Learning in the Presence of Concept Drift (2010)	2010	1.0	1.0	1.0	1.0	**4.0**	≥1.5	**Include**
9	Unsupervised Feature Selection for Outlier Detection on Streaming Data to Enhance Network Security (2021)	2021	0.0	1.0	1.0	0.0	**2.0**	≥2.5	**Exclude**
10	An online ensemble classification algorithm for multi-class imbalanced data stream (2024)	2024	0.0	1.0	1.0	0.0	**2.0**	≥2.5	**Exclude**
11	Active Weighted Aging Ensemble for drifted data stream classification (2023)	2023	0.0	1.0	1.0	0.0	**2.0**	≥2.5	**Exclude**
12	A Four-stage Bayesian Incremental Learning Framework for Intrusion Detection in Real-world IoT Environment (2026)	2026	0.5	0.5	1.0	0.0	**2.0**	≥2.5	**Exclude**

This retained set of 8 articles constitutes the eligible pool for Section 5.1 (Online and Incremental Learners) and forms part of the input to the synthesis and thematic analysis stage.

4. Data Integrity Notes

Each article was scored against the four quality criteria on the basis of its full text, applying the review-specific interpretation of QC1 and QC4 set out in Section 1 above. The retain or exclude decision was applied strictly according to the threshold in effect for that article's publication year, namely 2.5 out of 4 for articles published 2020 or later, or 1.5 out of 4 for articles published before 2020. No article was retained or excluded on any basis other than its quality score relative to its applicable threshold.

5. Document Preparation

This report has been prepared in accordance with the quality assurance procedure prior to proceeding to the synthesis stage.

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